Institutional Memory: Unified Multi-Source Knowledge Graph & AI Analytics for CNC Systems

A Neuro-Symbolic Architecture Bridging High-Frequency DAQ Telemetry, Technical Documentation, and Organizational Communications

Project: CNC DAQ Institutional Memory	**Stack:** Python / FastAPI / Neo4j / React / Qdrant
Core Thesis: Grounded Reasoning & Deterministic Physics	**Verification:** 135 Backend Tests, 50 UI Vitests (100% Pass)

1. Executive Summary & Research Motivation

Modern manufacturing laboratories generate heterogeneous data across 3 distinct silos: **(1) high-frequency sensor DAQ** capturing cutting forces and vibrations at 10 kHz, **(2) technical documentation** defining operating boundaries and ISO standards, and **(3) human communications (emails)** documenting maintenance logs, calibration notices, and anomalous incidents. Combining these streams manually requires deep tribal knowledge, causing severe institutional memory loss when personnel depart.

This project delivers an end-to-end **Neuro-Symbolic Institutional Memory System**. The system fuses telemetry from 822 CNC milling experiments across 25 process conditions, 7 engineering PDFs, and machine-relevant operator emails into an interconnected **Knowledge Graph** and hybrid vector retrieval index.

> ***Core Architectural Principle:*** *"The AI Explains, It Never Calculates."*
> *LLMs hallucinate arithmetic and signal math. In this system, all signal preprocessing, windowing, FFT extraction, baseline computation, and Mahalanobis anomaly detection are 100% deterministic code. The LLM receives strictly computed findings and is gated from fabricating numeric values.*

2. Unified Multi-Modal Ontology & Knowledge Graph

The graph store is structured around 18 strongly-typed entity classes and 20 directional relationship types with cryptographic provenance tracking.

Entity Category	Node Types	Key Relationships
Machine & DAQ	Condition, Session, Recording, SignalWindow, Finding, Channel	CAPTURED_UNDER, HAS_WINDOW, EXHIBITS_PATTERN, SIMILAR_TO
Physical Domain	Component, Fault, Process, Parameter, Procedure, Concept	PART_OF, CAUSED_BY, REQUIRES, EXPLAINS
Documentation	Document, Standard, GlossaryTerm	DOCUMENTS, SUPERSEDES, MENTIONED_IN
Human Context	Person, Organization, Email, EmailThread	SENT, RECEIVED, MENTIONS, CORROBORATES

3. The 6-Step Unified Query Execution Pipeline

User questions execute through a deterministic multi-stage neuro-symbolic pipeline:

Step	Stage Name	Mechanism & Deterministic Implementation
Step 1	**Intent & Entity Understanding**	Ontology NER, Channel Normalization, Sequential Session Resolving (1..822), and seed node extraction.
Step 2	**Strict CNC Scope Gate**	3-Tier safety gate (Entity match → Domain heuristic → 1-token LLM). Irrelevant questions are declined with IN_SCOPE_DECLARE verbatim with zero generation cost.
Step 3	**Multi-Hop Graph Traversal**	Bounded BFS exploration (max 2 hops, 50 nodes, 100 edges) expanding across People ↔ Emails ↔ Components ↔ Sessions ↔ Findings.
Step 4	**Node-Guided Data Retrieval**	Pulls underlying evidence per traversed node: full email bodies, PDF technical passages, computed session findings, and glossary definitions.
Step 5	**Deterministic Analytics & Reasoning**	On-demand session anomaly computation if uncomputed; caches results without redundant passes; structures cross-source correlation links.
Step 6	**Grounded Generation & Citations**	Synthesizes multi-source evidence prompt with mandatory distinct citation badges ([Finding N], [Doc N], [Email N]).

4. Deterministic Signal Processing & Anomaly Physics

- **Sensor Channels:** Spindle Vibration (Vib_Spindle), Table Vibration (Vib_Table), 3-Axis Force (X_Load_Cell, Y_Load_Cell, Z_Load_Cell), Motor Current (Current).
- **Sampling & Windowing:** Measured 10,000 Hz sample rate verified against time deltas; partitioned into 2,048-sample fixed windows with 50% overlap.
- **Feature Extraction:** 9 orthogonal features: Mean, Std Dev, RMS, Peak, Crest Factor, Excess Kurtosis, Skewness, Band Energy, Spectral Centroid.
- **Condition Baselines & Mahalanobis Distance:** Per-condition normal profiles are built exclusively from canonical recordings. Anomaly scores are computed via diagonal Mahalanobis distance with sigma departure rankings.
- **CSV Quality Gate:** Checksumming, null-byte sanitization, and canonical schema enforcement. Corrupt/truncated CSVs are isolated into quarantine.

5. Machine Email Relevance Quality Filter

- **Domain Keyword & Parameter Matching:** Strictly matches authentic CNC hardware terms (spindle, cutter, carbide insert, dynamometer, bearings), DAQ channel tokens, process conditions (S, F, D), or quality standards (ISO 3685, flank wear VB).
- **Automated Graph Extraction:** Accepted emails create Email nodes and link Person, Component, Fault, and Condition nodes via typed edges.
- **Negative Quality Discard:** Non-machine emails (e.g., Enron trading, general administration, payroll) are discarded with logged justifications and never indexed.

6. System Architecture & Dataset-Free Execution

A key design capability is running in **portable/dataset-independent mode** without copying raw multi-gigabyte CSVs:

Folder / Path	Size	Role & Runtime Necessity
knowledge_docs/	~21 MB	**Required.** Contains Tier-1 PDF manuals, engineering glossaries, machine fact sheets, and machine-relevant .eml emails.
artifacts/	~436 KB	**Required.** Stores trained offline tool-wear ML regression models (model.joblib) and performance metrics.
runtime/daq_cache/	~50-100 MB	**Required for instant startup.** Pre-computed analysis JSON files (windows, extracted features, baselines, and findings) for all 25 conditions.
config/	~12 KB	**Required.** domain.yaml holding machine ontology, channel mappings, condition matrices, and sensor thresholds.
dataset/Readngs_DAQ_Data/	~2-5 GB	**Optional.** Raw sensor CSVs. Only needed if re-computing fresh signal windows from scratch. The system runs 100% of features from cache.

7. Test Verification & Empirical Results

- **Backend Test Suite: 135 passing automated pytest test cases** (2 skipped live-data assertions), verifying windowing, quality quarantining, ontology traversal, scope gating, session caching, and email grounding.
- **Frontend Test Suite: 50 passing Vitest UI component tests** validating interactive signal waveforms, chat streaming, citation chip rendering ([S1], [E1], [F1]), and config diffs.
- **Tool Wear Regression Model:** Supervised ExtraTrees regressor predicting continuous flank wear (VB) with $R^2 = 0.9316$, $MAE = 14.32\,\mu\text{m}$, and $RMSE = 18.91\,\mu\text{m}$.
- **Grounded Citation Integrity:** Answers citing findings, documents, and emails link directly to exact source lines and window sample indices, eliminating hallucination.

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